Core java

Loops, bitwise , conditional statements

```
public class bitwise {
    public static void main(String[] args) {
        System.out.println("Hello bitwise");

        // bitwise operator  => AND, OR, XOR, ~, >>, <<

        int a= 3;
        int b= 5;

        System.out.println(a>>1);
        System.out.println(a<<2);
        // 0000 -> 0
        // 0010 -> 1
        // 0100 -> 2
        // 0110 -> 3
        // 1000 -> 4
        // 1010 -> 5
        // 1100 -> 6
        // 1110 -> 7
        System.out.println(a & b);
        System.out.println(a | b);
        // 0110
        // 1010
        // ----
        // 1110
        System.out.println(a ^ b);
        // 0110
        // 1010
        // ----
        // 1100
        // strict or either true or false
        System.out.println(~a);
        // 0110 -> 1001
        // 0111 -> 1000


        // ternary operator

        boolean isEligible = false;

        String value = isEligible ? "Eligible to vote" : "not eligible to
vote";

        System.out.println(value);
```

```java
        // conditional statement

        // controls flow of the program

        // if, if else, else if , nested if, switch case

        if(true){
            System.out.println("statement is true");
        }
        if(true){
            System.out.println("true");
        }else{
            System.out.println("False");
        }

       // nested if

        int age = 10;
        boolean voterId=true;

        if(age>=18){
            if(voterId){
                System.out.println("you are eligible for vote");
            }else{
                System.out.println("you don't have voter id ");
            }
        }else{
            System.out.println("you are not eligible to vote your age is less
then 18 ");
        }

        // else if ladder

        int mark = 80;
        if(mark >= 90){
            System.out.println("O");
        }else if (mark >=80){
            System.out.println("A");
        }else if(mark >= 70){
            System.out.println("B");
        }else if (mark >= 40){
            System.out.println("Just pass");
        }else{
            System.out.println("Fail");
        }

        // switch case
        // String name ="fiit";
```

```java
switch(mark){
    // case "academy":System.out.println("academy");
    // break;

    // case "fiit":System.out.println("FIIT");
    // break;
    case 90: System.out.println("O");
    break;

    case 80:System.out.println("A");
    break;

    case 50: System.out.println("pass");
    break;

    default:System.out.println("Fail");
    break;
}
// loops

// for , while, do-while

// System.out.println(1);
// System.out.println(2);
// System.out.println(3);
// System.out.println(4);

// for loop
// for(initilization, condition, increment/decrement){
    // block of code
// }

// for(int i=1; i<=10; i++){
//     System.out.println(i);
// }

// task to print only even number from 1 to 100

// for(int i=2; i<=100; i+=2){
//     System.out.println(i);
// }

// while loop


// 1 <=10 => true;
```

```
        // while(check <=10){
        //     System.out.println(check++);
        //     // check++;
        // }

        //  while(mark>=90){
        //     System.out.println(mark);
        // }

        // do {
        //     System.out.println("your grade is A");
        //     mark--;

        // } while (mark>=90);
        // 80 >= 90 -> false

    }

}
```

Theory Notes

1. Bitwise Operators

- Work directly on binary representation of numbers.
- Common operators:
  - & (AND) → 1 if both bits are 1.
  - | (OR) → 1 if at least one bit is 1.
  - ^ (XOR) → 1 if bits are different.
  - ~ (NOT) → flips all bits.
  - << (left shift) → shifts bits left (multiply by 2).
  - >> (right shift) → shifts bits right (divide by 2).

2. Ternary Operator

- Shorthand for if-else.

String result = (age >= 18) ? "Eligible" : "Not Eligible";

3. Conditional Statements

- **if** → executes block if condition is true.

- **if-else** → executes one block if true, another if false.
- **else-if ladder** → multiple conditions checked sequentially.
- **nested if** → if inside another if.
- **switch** → cleaner alternative for multiple discrete values.

4. Loops

- **for loop** → used when iteration count is known.
- **while loop** → runs until condition is false.
- **do-while loop** → runs at least once, then checks condition.
- Control statements:
    - break → exits loop immediately.
    - continue → skips current iteration.

Practice Tasks

Short Answer

1. What is the difference between & and && in Java?
2. How does << differ from >>?
3. Why does a do-while loop always execute at least once?
4. When should you use a switch instead of an if-else ladder?

Coding Exercises

1. Write a program to demonstrate all bitwise operators with two integers.
2. Use a ternary operator to check if a number is even or odd.
3. Implement a grading system using if-else ladder.
4. Create a nested if to check voting eligibility (age ≥ 18 and has voter ID).
5. Write a switch-case program to print the day of the week based on a number.
6. Print all even numbers from 1 to 100 using a for loop.
7. Demonstrate a do-while loop that prints a message at least once even if condition is false.